



AJKD Atlas of Renal Pathology: Minimal Change Disease

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Clinical and Pathologic Features

Minimal change disease (MCD) is characterized by nephrotic syndrome. It is the most common cause of nephrotic syndrome in children aged 1 to 7 years and remains a cause of nephrotic syndrome in adults.

Kidney function is usually preserved, except for occasional cases with acute kidney injury.

MCD is typically responsive to steroids with excellent long-term prognosis.

Light microscopy: Unremarkable glomeruli. Of note, global glomerulosclerosis is a nonspecific finding that can be seen.

Immunofluorescence microscopy: No deposits (may have nonspecific immunoglobulin M [IgM]).

Electron microscopy: Extensive foot process effacement.

Etiology/Pathogenesis

Abnormal cytokines from T cells and/or abnormal T-cell regulation by B cells are implicated. These may cause increased CD80 on podocytes in MCD.

MCD may also occur secondary to lymphomas, with hypersensitivity reactions (eg, acute interstitial nephritis from nonsteroidal anti-inflammatory drugs), with HIV (human immunodeficiency virus) infection, or with bee stings. In addition, extensive foot process effacement with nephrotic syndrome has been observed in patients with mild mesangial lupus nephritis and mild IgA nephropathy and may represent MCD-like injury in those patients.

Differential Diagnosis

Unsampled focal segmental glomerulosclerosis (FSGS); an adequate sample (ideally >25 glomeruli, including juxtamedullary area) is needed to reasonably exclude FSGS.

Degree of foot process effacement does not differentiate between unsampled primary FSGS and MCD.

Limited foot process effacement (<50%) suggests the process is probably not MCD or primary FSGS, unless the patient has been treated with partial response.

CD44 expression on visceral epithelial cells may indicate activated parietal epithelial cells transitioning to the tuft and could be a sign of early FSGS-type injury.

Key Diagnostic Features

- No immune complexes
- No segmental sclerosis
- Extensive foot process effacement

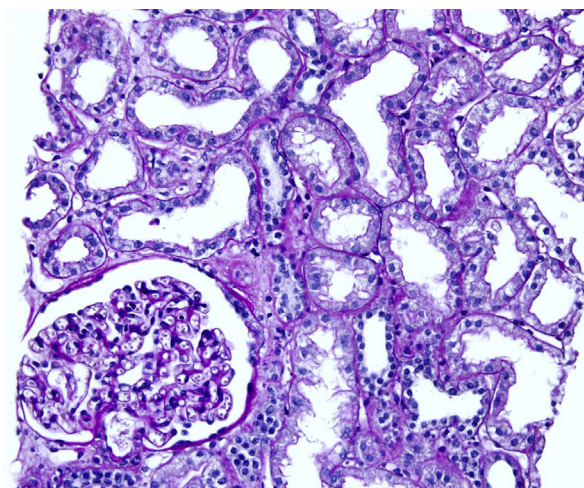


Figure 1. Preserved cortex with normal-appearing glomerulus. In this case, the tubules are distended with loss of brush border (periodic acid–Schiff stain).

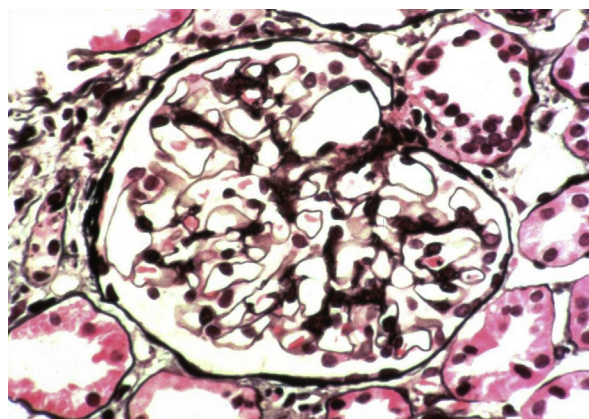


Figure 2. The glomerulus shows no proliferation, sclerosis, inflammation, or necrosis, and Bowman space is unremarkable. The capillary walls are delicate and of normal thickness (Jones silver stain). There are no deposits by immunofluorescence (not shown). Reproduced with permission from AJKD 33(3):e1.

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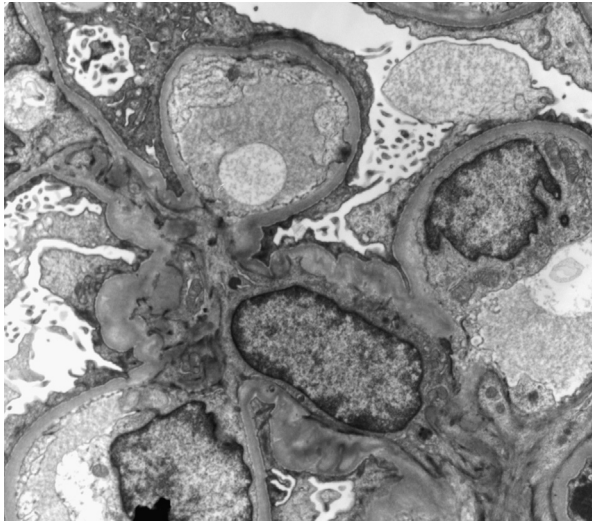


Figure 3. Diffuse effacement of foot processes of podocytes, with no immune complexes, normal mesangial cellularity and matrix, and normal thickness of the glomerular basement membrane (electron microscopy).

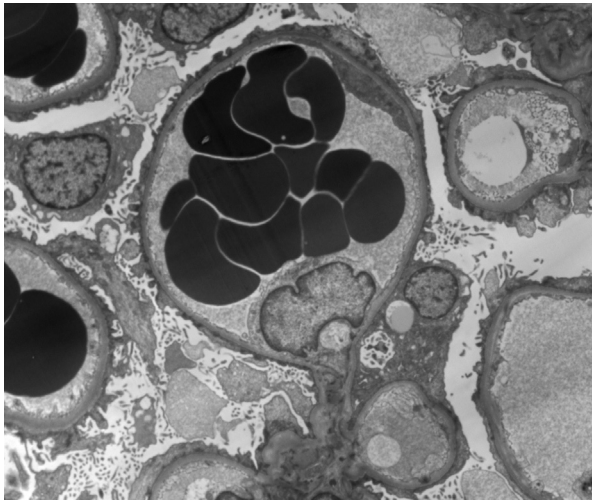


Figure 4. The injury to the podocytes is evident, with cytoplasmic vacuoles and microvillous transformation (electron microscopy).

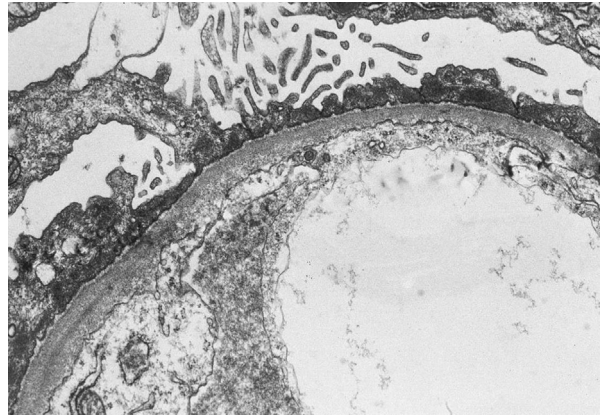


Figure 5. The surface of the injured podocytes becomes ruffled, and cytoplasmic extrusions are seen, so-called microvillous transformation, in addition to the effacement of foot process with loss of slit diaphragms (electron microscopy). Reproduced with permission from *AJKD* 33(3):e1.

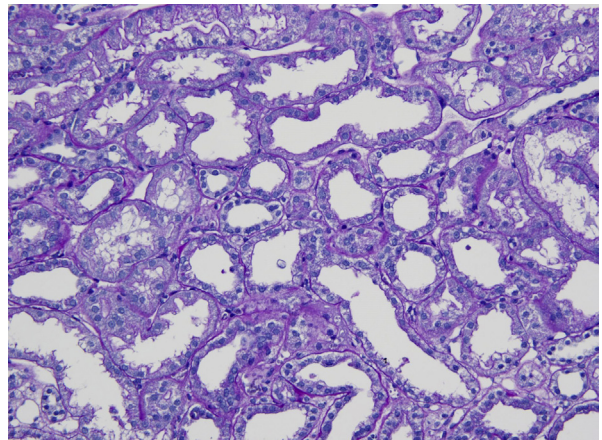


Figure 6. In some patients with minimal change disease, there may be accompanying tubular injury, as seen here (periodic acid-Schiff stain).